

NAME

OpenFECalculateAbsoluteBindingFreeEnergy.py - Calculate absolute binding free energy

SYNOPSIS

```
OpenFECalculateAbsoluteBindingFreeEnergy.py [--abfeParams <Name,Value,...>] [--executeDAGParams
<Name,Value,...>] [--loggingLevel <Info, Warning or Error>] [--mode <FirstMolecule, AllMolecules, or ...>] [
--missingChargeMode <Calculate or Stop>] [--moleculeNames <MolName1,MolName2,...>] [--outfilePrefix
<text>] [--overwrite] [--resultFileParams <Name,Value,...>] [--solventParams <Name,Value,...>] [
--smallMolFileParams <Name,Value,...>] [-w <dir>] -i <infile> -s <smallmolfile> -o <outfiledir>
```

```
OpenFECalculateAbsoluteBindingFreeEnergy.py -l | --list
```

```
OpenFECalculateAbsoluteBindingFreeEnergy.py -h | --help | -e | --examples
```

DESCRIPTION

Calculate Absolute Binding Free Energy (ABFE) for molecules in a small molecule input file. You may calculate ABFEs for specific molecules or all molecules in the input file.

The input file must contain a macromolecule already prepared for simulation. The preparation of the macromolecule for a simulation generally involves the following tasks: identification and replacement of non-standard residues; addition of missing residues; addition of missing heavy atoms; addition of missing hydrogens.

In addition, the small molecule input file must contain molecules already prepared for simulation. It must contain appropriate 3D coordinates relative to the macromolecule along with no missing hydrogens.

The MD simulation workflow, employed for the calculation of ABFEs, involves the following steps:

Protocol repeats, 3

Time step size: 4.0 femtosecond

Complex equilibration phase:

```
Max minimization steps: 5,000
NVT equilibration length: 0.25 nanosecond
NPT equilibration length: 0.5 nanosecond
NPT length: 5.0 nanosecond
```

Complex production phase:

```
Max minimization steps: 5,000
NPT equilibration length: 1.0 nanosecond
NPT length: 10.0 nanosecond
```

Solvent equilibration phase:

```
Max minimization steps: 5,000
NVT equilibration length: 0.1 nanosecond
NPT equilibration length: 0.2 nanosecond
NPT length: 0.5 nanosecond
```

Solvent production phase:

```
Max minimization steps: 5,000
NPT equilibration length: 1.0 nanosecond
NPT length: 10.0 nanosecond
```

Each complex and solvent simulation, by default, may run for 16.75 and 11.8 nanosecond respectively, for a total of 28.55 nanoseconds. The total MD simulation time for correspond to 85.66 nanosecond to repeat the protocol 3 times for the complex and solvent simulations.

Possible outfile prefix:

<OutfilePrefix> or <SmallMolFileRoot>

Possible output directories:

<OutfileDir>

<OutfileDir>/Transformations

<OutfileDir>/Results

Possible output files and directories under <OutfileDir>:

<OutfilePrefix>_ABFE_Results.<csv or tsv>

Transformations/<MolName>_AbsoluteBinding.json

... ..

Results/<MolName>_AbsoluteBinding_Results.json

Results/shared_AbsoluteBindingComplexUnit-*/

Results/shared_AbsoluteBindingSolventUnit-*/

... ..

OPTIONS

-a, --abfeParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for ABFE protocol settings employed during the calculation of ABFEs.

The default values are automatically updated to match settings provided by OpenFE module AbsoluteBindingProtocol.

You must specify valid OpenFE values for these parameters. An extensive validation is not performed.

The supported parameter names along with their default values are shown below: explain and doc...

protocolRepeats, 3

Complex equil output settings:

```
complexEquilOutputCheckpointInterval, 1 [ Units: nanosecond ]
complexEquilOutputCheckpointStorageFilename, checkpoint.chk
complexEquilOutputEquilNPTStructure, equil_npt_structure.pdb
complexEquilOutputEquilNVTstructure, equil_nvt_structure.pdb
complexEquilOutputForcefieldCache, db.json
complexEquilOutputLogOutput, production_equil_simulation.log
complexEquilOutputMinimizedStructure, minimized.pdb
complexEquilOutputIndices, all [ Possible value: Any valid
selection. ]
complexEquilOutputPremnimizedStructure, system.pdb
complexEquilOutputProductionTrajectoryFilename, production_equil.xtc
complexEquilOutputTrajectoryWriteInterval, 20.0 [ Units:
picosecond ]
```

Complex equil simulation settings:

```
complexEquilSimulationEquilibrationLength, 0.5 [ Units: nanosecond ]
complexEquilSimulationEquilibrationLengthNVT, 0.25 [ Units:
nanosecond ]
complexEquilSimulationMinimizationSteps, 5000
complexEquilSimulationProductionLength, 5.0 [ Units: nanosecond ]
```

Complex lambda settings:

```
complexLambdaElec, 0.0 0.0 0.0 0.0 0.0 0.0 0.1 0.2 0.3 0.4 0.5 0.6
0.7 0.8 0.9 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0
1.0 1.0 [ Possible values: A space delimited list of values
between 0.0 and 1.0 ]
complexLambdaRestrains, 0.0 0.2 0.4 0.6 0.8 1.0 1.0 1.0 1.0 1.0
1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0
1.0 1.0 1.0 1.0 [ Possible values: A space delimited list of
values between 0.0 and 1.0 ]
complexLambdaVdw, 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
0.0 0.0 0.0 0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.65 0.7 0.75 0.8 0.85
0.9 0.95 1.0 [ Possible values: A space delimited list of values
between 0.0 and 1.0 ]
```

Complex output settings:

```
complexOutputCheckpointInterval, 1.0 [ Units: nanosecond ]
complexOutputCheckpointStorageFilename, complex_checkpoint.nc
complexOutputForcefieldCache, db.json
complexOutputFilename, complex.nc
complexOutputIndices, not water [ Possible value: Any valid
selection. ]
complexOutputStructure, alchemical_system.pdb
complexOutputPositionsWriteFrequency, 100.0 [ Units: picosecond ]
complexOutputVelocitiesWriteFrequency, None [ Possible
values: > 0; Units: picosecond ]
```

Complex simulation settings:

```
complexSimulationEarlyTerminationTargetError, 0.0 [ Units:
kilocalorie_per_mole ]
complexSimulationEquilibrationLength, 1.0 [ Units: nanosecond ]
complexSimulationMinimizationSteps, 5000
complexSimulationNReplicas, 30
complexSimulationProductionLength, 10.0 [ Units: nanosecond ]
complexSimulationRealTimeAnalysisInterval, 250.0 [ Units:
picosecond ]
complexSimulationRealTimeAnalysisMinimumTime, 500.0 [ Units:
picosecond ]
complexSimulationSamplerMethod, repex [ Possible values: repex,
sams, or independent ]
complexSimulationSamsFlatnessCriteria, logZ-flatness [ Possible
values: logZ-flatness, minimum-visits or histogram-flatness ]
complexSimulationSamsGamma0, 1.0
complexSimulationTimePerIteration, 2.5 [ Units: picosecond ]
```

Complex solvation settings:

```
complexSolvationBoxShape, dodecahedron [ Possible values: cube,
dodecahedron, or octahedron ]
complexSolvationBoxSize, None [ Possible value: A triplet of space
X Y Z values; Units: nanometer ]
complexSolvationSolventModel, tip3p [ Possible values: tip3p, spce,
tip4pew, or tip5p ]
complexSolvationSolventPadding, 1.0 [ Units: nanometer ]
```

Engine settings:

```
engineComputePlatform, CPU [ Possible values: CPU, CUDA,
OpenCL, or Reference ]
engineGpuDeviceIndex, None [ Possible values: 0, 0 1, etc. ]
```

Forcefield settings:

```
forcefieldConstraints, HBonds [ Possible values: HBonds,
```

AllBonds, or HAngles]
 forcefields, amber/ff14SB.xml amber/tip3p_standard.xml
 amber/tip3p_HFE_multivalent.xml amber/phosaa10.xml
 [Possible values: A space delimited list of valid names.]
 forcefieldHydrogenMass, 3.0 [Units: amu]
 forcefieldNonbondedCutoff, 0.9 [Units: nanometer]
 forcefieldNonbondedMethod, PME [Possible values: PME or
 NoCutoff]
 forcefieldRigidWater, yes, [Possible values: yes or no]
 forcefieldSmallMoleculeForcefield, openff-2.1.1 [Possible
 value: A valid forcefield name.]

Integrator settings:

integratorBarostatFrequency, 25.0 * timestep [The specified value
 is a multiple of integratorTimestep.]
 integratorConstraintTolerance, 1e-06
 integratorLangevinCollisionRate, 1.0 [Units: 1 / picosecond]
 integratorNRestartAttempts, 20
 integratorReassignVelocities, no [Possible values: yes or no]
 integratorRemoveCom, no [Possible values: yes or no]
 integratorTimestep, 4.0 [Units: femtosecond]

Partial charge settings:

partialChargeNaglModel, None [Default: Production AM1BCC model for
 NAGL; Possible value: Any valid name.]
 partialChargeNumberOfConformers, None [Possible value: > 0]
 partialChargeOffToolkitBackend, AmberTools [Possible values:
 AmberTools or RDKit]
 partialChargeMethod, AM1BCC [Possible values: AM1BCC, Espaloma,
 or NAGL]

Restraint settings:

restraintKPhiA, 334.72 [Units: kilojoule_per_mole / radian**2
 The default value is equivalent to 80 kcal/mol/radian**2]
 restraintKPhiB, 334.72 [Units: kilojoule_per_mole / radian**2
 The default value is equivalent to 80 kcal/mol/radian**2]
 restraintKPhiC, 334.72 [Units: kilojoule_per_mole / radian**2
 The default value is equivalent to 80 kcal/mol/radian**2]
 restraintKR, 4184.0 [Units: kilojoule_per_mole / nanometer**2
 The default value is equivalent to 10 kcal/mol/angstrom**2]
 restraintKThetaA, 334.72 [Units: kilojoule_per_mole / radian**2
 The default value is equivalent to 80 kcal/mol/radian**2]
 restraintKThetaB, 334.72 [Units: kilojoule_per_mole / radian**2
 The default value is equivalent to 80 kcal/mol/radian**2]
 restraintAnchorFindingStrategy, bonded [Possible values:
 multi-residue or bonded]
 restraintDsspFilter, yes [Possible values: yes or no]
 restraintHostMaxDistance, 1.5 [Units: nanometer]
 restraintHostMinDistance, 0.5 [Units: nanometer]
 restraintHostSelection, backbone [Possible value: Any valid
 selection.]
 restraintRmsfCutoff, 0.1 [Units: nanometer]

Solvent equil output settings:

solventEquilOutputCheckpointInterval, 1.0 [Units: nanosecond]
 solventEquilOutputCheckpointStorageFilename, checkpoint.chk
 solventEquilEquilOutputNPTStructure, equil_npt_structure.pdb
 solventEquilEquilNVTOutputStructure, equil_nvt_structure.pdb
 solventEquilOutputForcefieldCache, db.json
 solventEquilOutputLogOutput, production_equil_simulation.log

```
solventEquilOutputMinimizedStructure, minimized.pdb
solventEquilOutputIndices, all [ Possible value: Any valid
    selection. ]
solventEquilOutputPreminimizedStructure, system.pdb
solventEquilOutputProductionTrajectoryFilename, production_equil.xtc
solventEquilOutputTrajectoryWriteInterval, 20.0 [ Units:
    picosecond ]

Solvent_equil_simulation_settings:

solventEquilSimulationEquilibrationLength, 0.2 [ Units: nanosecond ]
solventEquilSimulationEquilibrationLengthNVT, 0.1 [ Units:
    nanosecond ]
solventEquilSimulationMinimizationSteps, 5000
solventEquilSimulationProductionLength, 0.5 [ Units: nanosecond ]

Solvent lambda settings:

solventLambdaElec, 0.0 0.25 0.5 0.75 1.0 1.0 1.0 1.0 1.0 1.0 1.0 1.0
    1.0 1.0 [ Possible values: A space delimited list of values
    between 0.0 and 1.0 ]
solventLambdaRestraints, 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
    0.0 0.0 0.0 [ Possible values: A space delimited list of values
    between 0.0 and 1.0 ]
solventLambdaVdw, 0.0 0.0 0.0 0.0 0.0 0.12 0.24 0.36 0.48 0.6 0.7
    0.77 0.85 1.0 [ Possible values: A space delimited list of values
    between 0.0 and 1.0 ]

Solvent output settings:

solventOutputCheckpointInterval, 1.0 [ Units: nanosecond ]
solventOutputCheckpointStorageFilename, solvent_checkpoint.nc
solventOutputForcefieldCache, db.json
solventOutputFilename, solvent.nc
solventOutputIndices, not water [ Possible value: Any valid
    selection. ]
solventOutputStructure, alchemical_system.pdb
solventOutputPositionsWriteFrequency, 100.0 [ Units: picosecond ]
solventOutputVelocitiesWriteFrequency, None [ Possible
    values: > 0; Units: picosecond ]

Solvent simulation settings:

solventSimulationEarlyTerminationTargetError, 0.0 [ Units:
    kilocalorie_per_mole ]
solventSimulationEquilibrationLength, 1.0 [ Units: nanosecond ]
solventSimulationMinimizationSteps, 5000
solventSimulationNReplicas, 14
solventSimulationProductionLength, 10.0 [ Units: nanosecond ]
solventSimulationRealTimeAnalysisInterval, 250.0 [ Unit: picosecond ]
solventSimulationRealTimeAnalysisMinimumTime, 500.0 [ Units:
    picosecond ]
solventSimulationSamplerMethod, repex [ Possible values: repex,
    sams, or independent ]
solventSimulationSamsFlatnessCriteria, logZ-flatness [ Possible
    values: logZ-flatness, minimum-visits or histogram-flatness ]
solventSimulationSamsGamma0, 1.0
solventSimulationTimePerIteration, 2.5 [ Units: picosecond ]

Solvent solvation settings:

solventSolvationBoxShape, dodecahedron [ Possible values: cube,
    dodecahedron, or octahedron ]
solventSolvationBoxSize, None [ Possible value: A triplet of space
```

X Y Z values; Units: nanometer]
 solventSolvationSolventModel, tip3p [Possible values: tip3p, spce,
 tip4pew, or tip5p]
 solventSolvationSolventPadding, 1.5 [Units: nanometer]

Thermo settings:

thermoPh, None [Possible values: > 0]
 thermoPressure, 1.0 [Units: bar]
 thermoRedoxPotential, None [Possible values: A valid float.
 Units: millivolts (mV)]
 thermoTemperature, 298.15 [Units: kelvin]

A brief description of parameters, taken from OpenFE documentation, is provided below:

protocolRepeats: Number of completely independent repeats of the
 entire sampling process.

Complex settings:

Complex parameters for the system, including the solvent model and
 the solvent padding.

Complex equil output settings:

Parameters controlling simulation output during equilibration
 phase of complex transformation.

complexEquilOutputCheckpointInterval: Frequency to write the
 checkpoint file.
 complexEquilOutputCheckpointStorageFilename: Checkpoint filename.
 complexEquilOutputEquilNPTStructure: NPT structure filename.
 complexEquilOutputEquilNVTstructure: NVT structure filename.
 complexEquilOutputForcefieldCache: Filename for caching small
 molecule residue templates.
 complexEquilOutputLogOutput: Simulation log filename.
 complexEquilOutputMinimizedStructure: Minimized structure filename.
 complexEquilOutputIndices: Selection string for selecting
 coordinates to write.
 complexEquilOutputPremnimizedStructure: Initial structure filename.
 complexEquilOutputProductionTrajectoryFilename: Trajectory filename.
 complexEquilOutputTrajectoryWriteInterval: Frequency for writing
 velocities to trajectory file.

Complex equil simulation settings:

Parameters controlling simulation during equilibration phase of
 complex transformation.

complexEquilSimulationEquilibrationLength: Length of the NPT
 equilibration phase.
 complexEquilSimulationEquilibrationLengthNVT: Length of the NVT
 equilibration phase.
 complexEquilSimulationMinimizationSteps: Maximum number of
 minimization steps to perform.
 complexEquilSimulationProductionLength: Length of the NPT
 production phase.

Complex lambda settings:

Lambda protocol parameters for complex transformation.

complexLambdaElec: List of lambda values for electrostatics. The
 values of 0 and 1 imply state A and state B respectively.

complexLambdaRestrains: List of lambda values for restraints. The values of 0 and 1 imply state A and state B respectively.
complexLambdaVdw: List of lambda values for van der Waals. The values of 0 and 1 imply state A and state B respectively.

Complex output settings:

Parameters controlling simulation output during final phase of complex transformation.

complexOutputCheckpointInterval: Frequency to write the checkpoint file.
complexOutputCheckpointStorageFilename: Checkpoint filename.
complexOutputForcefieldCache: Filename for caching small molecule residue templates.
complexOutputFilename: Trajectory filename.
complexOutputIndices: Selection string for selecting coordinates to write.
complexOutputStructure: Topology structure filename.
complexOutputPositionsWriteFrequency: Frequency for writing positions to trajectory file.
complexOutputVelocitiesWriteFrequency: Frequency for writing velocities to trajectory file.

Complex simulation settings:

Parameters controlling simulation during final phase of complex transformation.

complexSimulationEarlyTerminationTargetError: Target error for the real time analysis measured in kcal/mol. Once the MBAR error of the free energy is at or below this value, the simulation will be considered complete. The suggested value of 0.12 has shown to be effective in both hydration and binding free energy benchmarks.
complexSimulationEquilibrationLength: Length of the equilibration phase. The specified value must be divisible by 'integratorTimestep'.
complexSimulationMinimizationSteps: Maximum number of minimization steps to perform.
complexSimulationNReplicas: Number of replicas to use.
complexSimulationProductionLength: Length of the production phase. The specified value must be divisible by 'integratorTimestep'.
complexSimulationRealTimeAnalysisMinimumTime: Time interval for performing analysis of the free energies. At each interval, real time analysis data will be written to a yaml file named <outputFileName>_real_time_analysis.yaml. The current error in the estimate will also be assessed and the simulation will be terminated when it drops below 'complexSimulationEarlyTerminationTargetError'.
complexSimulationSamplerMethod: Alchemical sampling method to use: REPEX (Hamiltonian REplica EXchange), SAMS (Self-Adjusted Mixture Sampling), or Independent (Independently sampled lambda windows).
complexSimulationSamsFlatnessCriteria: Method for assessing when to switch to asymptotically optimal scheme for SAMS.
complexSimulationSamsGamma0: Initial weight adaptation rate for SAMS.
complexSimulationTimePerIteration: Simulation time between each MCMC move attempt

Complex solvation settings:

Solvation parameters for the system, including the solvent model and

the solvent padding.

complexSolvationBoxShape: Shape of the periodic solvent box.
complexSolvationBoxSize: Lengths of the unit cell for a solvent box.
complexSolvationSolventModel: Forcefield water model to use during solvation and defining the model properties.
complexSolvationSolventPadding: Minimum distance from any solute bounding sphere to the edge of the box.

Engine settings:

Parameters configuring the compute platform used by the OpenMM to perform the simulation.

engineComputePlatform: Platform to use for running OpenMM MD calculations.
engineGpuDeviceIndex: Space delimited list of device indices to use for running OpenMM MD calculations.

Forcefield settings:

forcefieldConstraints: Constraints to use.
forcefields: List of valid forcefield paths for all components except small molecules.
forcefieldHydrogenMass: Mass to be repartitioned to hydrogens from neighboring heavy atoms.
forcefieldNonbondedCutoff: Cutoff for short range nonbonded interactions.
forcefieldNonbondedMethod: Method for treating nonbonded interactions.
forcefieldRigidWater: Use a rigid water model.
forcefieldSmallMoleculeForcefield: A valid forcefield name to use for small molecules.

Integrator settings:

Parameters controlling the LangevinSplittingDynamicsMove integrator used for simulation.

integratorBarostatFrequency: Frequency at which volume scaling changes should be attempted.
integratorConstraintTolerance: Tolerance for constraint solver.
integratorLangevinCollisionRate: Collision frequency.
integratorNRestartAttempts: Number of attempts to restart from Context in case there are NaNs in the energies after integration.
integratorReassignVelocities: Reassign velocities from the Maxwell-Boltzmann distribution at the beginning of each Monte Carlo move.
integratorRemoveCom: Remove the center of mass motion.
integratorTimestep: Size of the simulation timestep.

Partial charge settings:

Parameters for automatically assigning missing partial charges to small molecules, including the partial charge method.

partialChargeNaglModel: Model to use for partial charge assignment. A value of None implies the use of the latest available production AM1BCC model.
partialChargeNumberOfConformers: Number of conformers to generate as part of the partial charge assignment. A value of None implies the use of the existing conformer.
partialChargeOffToolkitBackend: OpenFF toolkit registry backend to

use for calculating partial charges.
 partialChargeMethod: Method to use for calculating partial charges.

Restraint settings:

Parameters to configure Boresch-style restraint between two groups of atoms named host (Hx) and guest (Gx).

restraintKPhiA: Equilibrium force constant for the dihedral formed by H2-H1-H0-G0.
 restraintKPhiB: Equilibrium force constant for the dihedral formed by H1-H0-G0-G1.
 restraintKPhiC: Equilibrium force constant for the dihedral formed by H0-G0-G1-G2.
 restraintKR: Bond spring constant between H0 and G0.
 restraintKThetaA: Spring constant for the angle formed by H1-H0-G0.
 restraintKThetaB: Spring constant for the angle formed by H0-G0-G1
 restraintAnchorFindingStrategy: Boresch atom picking strategy to use.
 bonded: pick host atoms that are bonded to each other.
 multi-residue: pick host atoms which can span multiple residues.
 restraintDsspFilter: Apply DSSP filter to the host atoms.
 restraintHostMaxDistance: Minimum distance between any host atom and the guest G0 atom.
 restraintHostMinDistance: Xaximum distance between any host atom and the guest G0 atom
 restraintHostSelection: A valid selection string to sub-select the host atoms which will be involved in the restraint.
 restraintRmsfCutoff: Cutoff value for filtering atoms by their root mean square fluctuation. Atoms with values above this cutoff are ignored.

Solvent equil output settings:
 Solvent equil simulation settings:
 Solvent lambda settings:
 Solvent output settings:
 Solvent simulation settings:
 Solvent solvation settings:

The solvent settings are similar to the complex settings already described under various sections for complex. The prefix 'solvent' is used for the names of the pramaters instead of the prefix 'complex.'

Thermo settings:

Thermodynamic parameters, including the temperature and the pressure of the system.

thermoPh: Simulation pH
 thermoPressure: Simulation pressure.
 thermoRedoxPotential: Simulation redox potential.
 thermoTemperature: Simulation temperature.

-e, --examples

Print examples.

--executedAGParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for executing protocol DAGs (Directed Acyclic Graph) to run ABFE calculations.

The supported parameter names along with their default values are are shown below:

keepShared, yes [Possible values: yes or no]
 keepScratch, no [Possible values: yes or no]

nRetries, 2 [Possible values: >= 0. A value of 0 implies only
1 try.]

A brief description of parameters is provided below:

keepShared: Keep shared directories after the execution of DAG.
keepScratch: Keep scratch directories after the execution of DAG.
nRetries: Number of times to attempt the execution.

-h, --help

Print this help message.

-i, --infile <infile>

Input file name containing a macromolecule.

-l, --list

List default ABFE protocol settings provided by OpenFE module AbsoluteBindingProtocol.

--loggingLevel <Info, Warning or Error> [default: Error]

Logging level to configure the 'root logger' via logging.basicConfig() function. The default logging level is changed from 'logging.INFO' to 'logging.ERROR'. Otherwise, OpenFE and its associated modules may generate a lot of informational messages.

-m, --mode <FirstMolecule, AllMolecules, or ...> [default: FirstMolecule]

Calculate ABFE for the first molecule, the specified molecule names, or all molecules in an input file. Possible values: FirstMolecule, AllMolecules, or MoleculeNames. You must specify a comma delimited list of molecule names using '--moleculeNames' option during 'MoleculeNames' value for '--mode' option.

--missingChargeMode <Calculate or Stop> [default: Stop]

Calculate missing partial charges for molecules before running ABFE calculations or terminate the execution of the script. The missing partial charges will be automatically calculated by OpenFE module AbsoluteBindingProtocol during the calculation of ABFE. You may control the calculation of partial charges by specifying values for partialCharge* parameters using '--abfe' option.

--moleculeNames <MolName1,MolName2,...>

A comma delimited list of molecule names for calculating ABFEs. This option is only used during 'MoleculeNames' value for '--mode' option.

-o, --outdir <outdir>

Output directory.

--outfilePrefix <text> [default: auto]

Prefix for generating output files under output directory.

--overwrite

Overwrite existing files.

--resultFileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for writing calculated RHFES values to a results file.

The supported parameter names along with their default values are shown below:

precision, 4 [Possible values: > 0]
delimiter, comma [Possible values: comma or tab]

--solventParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for solvent component. You must specify valid OpenFE values. No extensive validation is performed. These parameters are used in conjunction with solvation* parameters available through '--abfeParams' to perform solvation.

The supported parameter names along with their default values are shown below:

```

positiveIon, Na+ [ Possible value: Li+, Na+, K+, Rb+, or Cs+ ]
negativeIon, Cl- [ Possible values: Cl-, Br-, F-, or I- ]
neutralize, yes [ Possible values: yes or no ]
ionConcentration, 0.15 [ Units: molar ]

```

A brief description of parameters is provided below:

```

positiveIon, negativeion: Pair of ions used to neutralize and bring
the solvent to required ionic concentration.
neutralize: Neutralize the net charge on the chemical state by the
ions in the solvent component.
ionConcentration: Ionic concentration.

```

-s, --smallMolFile <SmallMolFile>

Input file containing small molecules.

--smallMolFileParams <Name,Value,...> [default: auto]

A comma delimited list of parameter name and value pairs for reading molecules from files. The supported parameter names for different file formats, along with their default values, are shown below:

```
SD: removeHydrogens,no, sanitize,yes, strictParsing,yes
```

-w, --workingdir <dir>

Location of working directory which defaults to the current directory.

EXAMPLES

The sample protein and ligand files for tyrosine kinase 2 (Tyk2) are distributed with MayaChemTools and are available in data directory. These files have been taken from OpenFE distribution for example notebooks. The AM1BCC partial charges have been calculated for the ligands in SD file to facilitate calculations. You may review OpenFE tutorial notebooks for the expected results.

To calculate ABFE for the first molecule in a SD file, performing 3 independent repeats of the entire MD sampling process to estimate ABFE for the molecule, each solvent and vacuum MD repeat consisting of equilibration phase (Complex: Minimization - 5,000; NVT - 0.25 ns; NPT - 0.5; NPT prod - 5.0 ns; Solvent: Minimization - 5,000; NVT - 0.1; NPT - 0.2 ns; NPT prod - 0.5 ns) and production phase (Complex: Minimization - 5,000; NPT equil - 1.0 ns; NPT prod: 10.0 ns; Solvent: Minimization - 5,000; NPT equil - 1.0 ns; NPT prod - 10 ns) using a step size of 4 fs, writing out appropriate trajectory and PDB files for each MD repeat in Results subdirectory under output directory, type:

```
% OpenFECalculateAbsoluteBindingFreeEnergy.py -i SampleTyk2.pdb
-s SampleTyk2Ligands.sdf -o SampleTyk2LigandsAHFE
```

To run the first example for calculating ABFE for specific molecules using CUDA platform on your machine to perform solvent and vacuum MD simulations and generate various output files, type:

```
% OpenFECalculateAbsoluteBindingFreeEnergy.py -i SampleTyk2.pdb
-s SampleTyk2Ligands.sdf -o SampleTyk2LigandsAHFE -m MoleculeNames
--moleculeNames "lig_ejm_31, lig_ejm_47"
--abfeParams "engineComputePlatform,CUDA"
```

To run the second example to see all warning messages produced by OpenFE modules and write various output files, type;

```
% OpenFECalculateAbsoluteBindingFreeEnergy.py -i SampleTyk2.pdb
-s SampleTyk2Ligands.sdf -o SampleTyk2LigandsAHFE -m MoleculeNames
--moleculeNames "lig_ejm_31, lig_ejm_47"
--abfeParams "engineComputePlatform,CUDA"
--loggingLevel Warning
```

To run the first example for calculating ABFE for all molecules using CUDA platform on your machine to perform solvent and vacuum MD simulations, automatically calculate missing partial charges for molecules, and generate various output files, type:

```
% OpenFECalculateAbsoluteBindingFreeEnergy.py -i SampleTyk2.pdb
-s SampleTyk2LigandsNoCharges.sdf -o SampleTyk2LigandsAHFE
-m AllMolecules --abfeParams "engineComputePlatform,CUDA"
--missingChargeMode Calculate
```

To run the second example by specifying explicit values for various parameters and generate various output files, type:

```
% OpenFECalculateAbsoluteBindingFreeEnergy.py -i SampleTyk2.pdb
-s SampleTyk2Ligands.sdf -o SampleTyk2LigandsAHFE -m MoleculeNames
--moleculeNames "lig_ejm_31, lig_ejm_47"
--loggingLevel Error
--executeDAGParams "keepShared, yes, nRetries, 2"
--missingChargeMode Stop --abfeParams "protocolRepeats,3,
engineComputePlatform,CUDA,integratorTimestep, 4.0,
complexEquilSimulationEquilibrationLengthNVT, 0.25,
complexEquilSimulationEquilibrationLength, 0.5,
complexEquilSimulationProductionLength,5.0,
complexSimulationEquilibrationLength, 1.0,
complexSimulationProductionLength, 10.0,
solventEquilSimulationEquilibrationLengthNVT, 0.1,
solventEquilSimulationEquilibrationLength, 0.2,
solventEquilSimulationProductionLength, 0.5,
solventSimulationEquilibrationLength, 1.0,
solventSimulationProductionLength, 10.0,
thermoPressure, 1.0, thermoTemperature, 298.15"
```

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SEE ALSO

OpenFECalculateAbsoluteHydrationFreeEnergy.py, OpenFECalculatePartialCharges.py,
OpenFECalculateRelativeBindingFreeEnergy.py, OpenFECalculateRelativeHydrationFreeEnergy.py,
OpenFEGenerateLigandNetwork.py

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The functionality available in this script is implemented using OpenFE, an open source molecular for alchemical free energy calculations.

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